TAO Wenwen

Curriculum Vitae — PhD and Graduate Admissions

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2027 PhD · Research Master · MPhil Applicant | Full Academic CV

B.M. Clinical Medicine, Nanchang University (2017–2022) | Chinese (Native) | English (IELTS 6.5, TOEFL 97)

Research Profile

Clinically trained multimodal human neuroscience researcher studying how dynamic brain, behavioral and physiological states relate to mental health and developmental variation. Patient-facing work first showed me how clinical heterogeneity complicates measurement; subsequent work in addiction cohorts taught me how recruitment, longitudinal follow-up, acquisition, quality control and data provenance determine whether a scientific question can ultimately be answered.

My later training in depression neuroimaging, repeated symptom measurement and neuromodulation deepened my experience in analysis and participant-level interpretation. I now work from the scientific question backward—independently designing synchronized multimodal research systems around downstream preprocessing, modeling and interpretation needs.

My current work focuses on youth mental health and the distinction between relatively stable vulnerability and within-person state change. I am particularly interested in translating clinically grounded measures into structured multimodal representations that can support computational modeling, individualized assessment and, ultimately, adaptive intervention.

Across these projects, I have increasingly worked across the full research chain: clinical question and protocol design → multimodal acquisition and synchronization → data quality and representation → computational analysis → translational application.

Research Interests

Youth Mental Health and Developmental Psychopathology · Computational Psychiatry · Multimodal Brain–Behavior Dynamics · Within-Person and Longitudinal Variation · Emotion Regulation and Cognitive Control · Multimodal Neuroimaging · Dynamic State Modeling · Neuromodulation · Translational Neuroscience

Education

B.M. in Clinical Medicine | Nanchang University | 2017–2022

GPA: 3.2/4.0 (82/100)

Relevant training: Psychiatry, Neurology, Internal Medicine, Surgery, Pathology, Pharmacology, Medical Statistics and Evidence-Based Medicine

Research Experience

Research Assistant | Tsinghua University, CogDrive Lab | Mar 2026–Present

State Key Laboratory of Intelligent Green Vehicle and Mobility

Advisor: Prof. Dongpu Cao

Clinical collaboration: Tongji Hospital, Huazhong University of Science and Technology

Multimodal developmental neuroscience of youth depression and translation to intelligent health-cockpit research

- **Scientific question:** Investigate how neural, hemodynamic, autonomic, behavioral and clinical signals characterize mental and emotional states across ages 6–18 while distinguishing developmental differences from clinically meaningful within-person and between-person variation.
- **Research ownership:** Independently translated this question into the overall study protocol, cognitive-affective paradigms and clinician-led interview workflow for a planned cohort of 300 clinical participants and 150 healthy controls. The protocol received joint scientific review and hospital IRB approval.
- **Methodological solution:** Designed the synchronized acquisition and event-trigger architecture integrating simultaneous 64-channel EEG–fNIRS, eye tracking, multi-view facial video, speech, ECG, PPG, GSR and respiration during computerized tasks and clinical interviews, with acquisition planned around downstream preprocessing, neurovascular-coupling and multimodal modeling requirements.
- Established an age-adaptive clinical ground-truth framework using CDRS-R, age-matched self-/parent-report measures, K-SADS-PL DSM-5 clinician confirmation and multisource severity grading, together with developmentally appropriate consent, assent and risk-escalation procedures.
- Mapped clinically meaningful behavioral anchors to observable multimodal features—including facial behavior, speech prosody, gaze patterns and physiological responses—to create interpretable links between psychiatric phenotypes and machine-readable signals.
- Operationalized structured psychiatric interview logic into age-adaptive workflows for four developmental bands (6–9, 9–12, 12–15 and 15–18 years), supporting standardized clinical interviewing while providing a structured representation that can later inform multimodal computational and AI-based assessment.
- Defined synchronization, metadata and quality-control standards across 13+ multimodal data streams, including temporal alignment at data, feature and semantic levels, standardized metadata conventions, graded QC labels and analysis-ready data specifications.
- Built BPlus MVP-0 to translate the study protocol into six executable modules (Rest, MMN, VFT, N-back, Emotion and Recovery), with participant-response capture, event/trigger logging, preflight checks and automated QC/state reports. Software demonstrations, dry runs and real-hardware synchronization testing were completed, and formal data collection has begun with healthy control participants.
- Independently coordinated and advanced the Tongji Hospital IRB process, translating the experimental protocol into ethics-ready procedures for minor protection, informed consent/assent, stigma-sensitive recruitment and risk escalation.
- Designed the translational pathway connecting laboratory-validated multimodal markers with lower-burden sensing and structured state representations for future intelligent health-cockpit and human–AI applications, while treating clinical validation and real-world deployment as later stages rather than established outcomes.

Research Intern | Tsinghua University, Yan Lab | Sep 2025–Mar 2026

Advisor: Prof. Chaogan Yan

Mind Flower depression research

Depression neuroimaging, repeated symptom measurement and individualized rTMS

- **Scientific question:** Examined how individual differences in brain function, cognition and repeated symptom trajectories could inform understanding of depression and treatment response, integrating clinical phenotyping, MRI, cognitive tasks, rTMS records and wearable follow-up.
- Conducted participant screening and independently administered MINI and serial HAMD/HAMA/MADRS assessments across repeated visits.
- Managed resting-state, rumination and distraction MRI sessions, including individualized negative-recall cue preparation, E-Prime task adaptation, head-motion monitoring (<3 mm) and OptiMRI audio quality control.
- Completed structured hands-on training in DPABI/DPABISurf and DeepHCP neuroimaging workflows and applied this training to resting-state and task-fMRI preprocessing, quality control and participant-level interpretation, extending beyond the basic QC work of earlier projects.
- Linked repeated HAMD-6 and rTMS tolerability records with WCST, SST and emotional n-back measures and two-week wearable monitoring of HRV, sleep and activity.
- Coordinated repeated experimental workflows across clinicians, counselors and participants and standardized E-Prime recovery/export, multimodal file naming, QC and hierarchical NAS archiving across longitudinal visits.

Visiting Research Intern | Beijing Normal University, Liu Lab | Sep 2025–Mar 2026
Advisor: Dr. Zheyun Liu
MRI-guided transcranial ultrasound stimulation of bilateral GPi in Parkinson's disease
- Supported individualized bilateral globus pallidus internus (GPi) targeting using structural MRI and Brainsight neuronavigation within a multidisciplinary transcranial ultrasound stimulation workflow.
- Operated an approximately 1.5-hour TUS–cognitive assessment protocol focused on memory and cognitive flexibility, documenting stimulation parameters, behavioral responses, safety and participant-reported auditory/thermal tolerability.
- Contributed to a standardized workflow for investigating whether non-invasive deep-brain ultrasound stimulation can modulate basal-ganglia-related cognitive function while maintaining consistent targeting, parameter recording and safety monitoring.

Research Assistant | Shenzhen Mental Health Center, Addiction Medicine | Jan 2024–Aug 2025
Advisors: Prof. Mei Yang, Prof. Xiaojian Jia, Dr. Gengdi Huang and Dr. Yi Zhang
Addiction neuroimaging and longitudinal clinical cohorts
- **Scientific question:** Investigated how clinical, circuit-level and biological factors relate to addiction severity and longitudinal outcomes in alcohol and methamphetamine use disorders.
- **Clinical-cohort responsibility:** Coordinated recruitment, screening, psychiatric assessment and longitudinal follow-up for 290+ participants. Worked with clinicians and department leadership to identify eligible participants and with radiology staff and technical engineers to prepare MRI/DTI, EEG, behavioral and biospecimen workflows; maintained >95% data completeness.
- Participated in resting-state and task-based MRI/DTI and EEG acquisition, quality control and basic preprocessing while building standardized participant, sample, metadata and anomaly-tracking procedures to support reproducible downstream analysis.
- Built experience in longitudinal clinical data management across MRI/DTI, EEG, behavioral assessment, 16S rRNA microbiome sequencing and GC–MS short-chain-fatty-acid metabolomics.
- This work established my foundation in running real clinical cohorts and showed me that acquisition decisions, participant communication, follow-up, data quality and provenance are inseparable from later modeling and interpretation.

Selected Research Projects and Contributions

Gut Microbiota–Metabolite–Circuit Mechanisms in AUD | 220 participants
Research question: Can microbiome and metabolic disruption be linked to white-matter injury and clinically relevant addiction phenotypes?
- Integrated 16S rRNA microbiome profiles, GC–MS short-chain-fatty-acid measures and multimodal MRI to examine a cross-scale gut–metabolite–brain pathway rather than treating each modality as an isolated endpoint.
- Identified three candidate microbial markers associated with SCFA levels and observed associations between lower SCFA concentrations and lower fractional anisotropy in the anterior limb of the internal capsule and corpus callosum.
- Treated these findings as preliminary cross-modal associations requiring longitudinal and mechanistic confirmation rather than causal evidence.

Neuroprotective Mechanisms of Tetramethylpyrazine in AUD | 160 participants
Research question: Does TMP intervention relate to change in white-matter microstructure?
- Coordinated multimodal imaging and assessed pre-/post-intervention FA in the corpus callosum and superior longitudinal fasciculus.
- Preliminary imaging differences require longitudinal confirmation.
- Optimized the combined DTI + resting-state fMRI acquisition and quality-control workflow, including procedures addressing susceptibility-related artifacts in subcortical imaging.

Methamphetamine-Related Frontotemporal Injury and Risky Decision-Making | 10 newly acquired cases + 80-case database
Research question: How do structural and task-state signals relate to cognitive impairment and risky choice in methamphetamine dependence?
- Independently acquired MRI and EEG data from 10 participants and maintained the broader clinical database.
- Contributed to DTI/fMRI analyses of prefrontal–hippocampal pathways, cingulum and fornix diffusion measures and distributed frontotemporal abnormalities.
- Led participant-held-out prediction analyses comparing task-state information with pre-decision EEG. The resulting first-author manuscript found dominant task-state effects and limited incremental predictive value from pre-decision EEG.

PFC–Ventral Striatal Executive-Control Network | 20 deeply phenotyped cases
Research question: Can structural–functional circuit measures help explain executive-control deficits and relapse vulnerability?
- Examined PFC–ventral striatal structural connectivity using DTI together with task-fMRI functional coupling and behavioral measures.
- Developed an exploratory logistic-regression model of relapse outcome.
- Used the findings to generate testable hypotheses for future connectivity-informed neuromodulation rather than presenting the exploratory model as a validated clinical predictor.

Publications and Manuscripts

- Tao, W. Early-Recovery Depressive Symptom Burden, First-Month Drinking, and Subsequent Alcohol Use After Inpatient Treatment: A Six-Month Prospective Cohort Study. Manuscript submitted to Drug and Alcohol Review. First author; prospective cohort and longitudinal outcome analysis.
- Tao, W. Temporal transportability and measurement stability of machine-learning models for post-discharge drinking in alcohol dependence: a single-centre retrospective evaluation. Submitted to BMC Medical Informatics and Decision Making. First author; temporal validation and comparative modeling.
- Tao, W. Within-person EEG decodability does not translate to cross-participant prediction of risky decisions in methamphetamine dependence. Submitted to International Journal of Psychophysiology. First author; participant-held-out EEG and behavioral prediction analysis.

Research Direction

- **Immediate empirical focus:** Use longitudinal, clinically grounded multimodal studies to distinguish relatively stable vulnerability from within-person state change in youth mental health and development.
- **Mechanistic focus:** Examine how control, reward, memory, affective and physiological systems dynamically coordinate across context, symptom state and developmental stage, including neurovascular coupling measured with simultaneous EEG–fNIRS.
- **Computational focus:** Develop interpretable representations of clinically meaningful internal states using multimodal neural, physiological, behavioral and longitudinal data rather than relying on isolated biomarkers.
- **Long-term translation:** Use state-sensitive models to inform individualized assessment and intervention, with adaptive neurotechnology and human–AI systems treated as later extensions of the same state-to-action scientific question rather than separate research identities.

Clinical Experience

Resident Physician | Wannian County Hospital of TCM | Dec 2022–Dec 2023

- Managed 15–20 chronic-disease inpatients through clinical assessment, treatment planning, inpatient management and follow-up, gaining practical experience in chronic-disease management and evaluating individual variation in real-world clinical settings.
- Adapted existing chronic-disease follow-up forms to local patient needs, adding structured longitudinal fields to track changes in symptoms, clinical status and treatment over time.

Clinical Intern | Ganzhou People's Hospital | Feb 2021–Feb 2022

- Completed 12 clinical rotations, including Neurology ICU and Cardiology ICU, with broad exposure to inpatient, emergency and perioperative care.
- Performed supervised lumbar puncture and ECG interpretation and participated in routine inpatient and surgical workflows.

Methods and Technical Expertise

Multimodal Experimental Design

Protocol and paradigm development; TTL/LSL and event-trigger architecture; synchronized acquisition; clinical interview workflow; multimodal QC; metadata design; analysis-ready data specifications

Neuroimaging and Connectivity Analysis

Structural MRI; resting-state and task fMRI; DTI tractography; FA/ADC; structural–functional connectivity; DeepHCP; DPABI/DPABISurf; FSL; individualized MRI-based targeting

Neurophysiology and Physiology

EEG (32/64-channel); simultaneous EEG–fNIRS; ECG; PPG; GSR; respiration; eye tracking; synchronized task/interview acquisition; signal QC; exploratory neurovascular-coupling analysis

Clinical and Longitudinal Research

MINI; K-SADS-PL DSM-5; CDRS-R; HAMD; HAMA; MADRS; PHQ-9; PHQ-15; CDI; SDQ; recruitment; repeated assessment; participant follow-up; cohort and IRB coordination; informed consent/assent; pediatric safeguards

Neuromodulation

rTMS; transcranial ultrasound stimulation; Brainsight neuronavigation; individualized MRI-based targeting; stimulation-parameter logging; tolerability and safety monitoring

Behavioral and Cognitive Paradigms

MMN; VFT; N-back; emotional n-back; WCST; SST; BART; Two-Step task; clinical interview paradigms; E-Prime and PsychoPy workflows

Quantitative and Computational Methods

Python; MATLAB; R; SPSS; regression/classification; cross-validation; AUC; participant-held-out validation; temporal validation; longitudinal and repeated-measures analysis

Clinical AI and Structured Representation

Multimodal data schemas; age-adaptive structured clinical dialogue workflows; clinically grounded input/output specifications; multimodal feature representation; exploratory RAG and workflow design for clinical AI applications

Microbiome, Metabolomics and Biospecimens

16S rRNA gut-microbiome sequencing; GC–MS short-chain-fatty-acid metabolomics; blood/stool/saliva processing; biospecimen collection, labeling and QC SOPs

Research Software and Data Infrastructure

BPlus MVP-0; participant-response and event logging; trigger architecture; automated preflight and QC reporting; longitudinal database design; Data Dictionary development; version control; NAS-based data management; multicenter harmonization

Languages

English — TOEFL 97; IELTS 6.5
Chinese — Native